

Sampling Techniques in Statistics

Jonathan R. Bacolod

Sampling is the process of selecting a smaller group from a larger **population** to gather information.

Population

A **population** is the complete set of people, objects, or events being studied.

Example: All Grade 7 students in a school.

Sample

A **sample** is a smaller part of the population chosen for investigation.

Example: 30 Grade 7 students chosen from the school.

Population vs. Sample

Population	Sample
Whole group	Part of the group
Larger set	Smaller set
Hard to study fully	Easier to study

Why Good Sampling Matters

Good sampling techniques help researchers:

- ▶ Collect information **efficiently**
- ▶ Represent the population **fairly**

A good sample acts like a *mini version* of the whole population.

Simple Random Sampling

Simple Random Sampling – every member has an equal chance of being selected.

Think: names drawn from a box.

Systematic Sampling

Systematic Sampling – members are selected at regular intervals after a random starting point.

Think: every 10th person in a line.

Stratified Sampling

Stratified Sampling – the population is divided into groups (strata), then samples are taken from each group.

Think: equal numbers from each grade level.

Cluster Sampling

Cluster Sampling – the population is divided into clusters, and one or more entire clusters are selected.

Think: surveying all households in a chosen barangay.

Convenience Sampling

Convenience Sampling – participants are chosen because they are easy to reach.

Think: asking the nearest classmates.

Quick Recap: 5 Sampling Techniques

Technique	Key Idea
Simple Random	Equal chance for all
Systematic	Regular intervals
Stratified	Groups, then sample each
Cluster	Whole clusters chosen
Convenience	Easiest to reach

Examples

Example 1

A librarian numbers 50 books and uses a computer to randomly pick 5 for inspection.

This is **Simple Random Sampling** because every book has an equal chance of being chosen.

Example 2

A guard checks every 20th car entering a parking lot after randomly choosing the first car.

This is **Systematic Sampling** because cars are chosen at regular intervals.

Example 3

A researcher selects 15 employees from each of 4 departments in a company.

This is **Stratified Sampling** because the population is divided into groups before sampling.

Example 4

A researcher randomly picks 2 out of 10 classrooms and surveys every student in those 2 classrooms.

This is **Cluster Sampling** because entire groups (classrooms) are selected.

Example 5

A vendor asks the first 10 people who pass by their stall to try a new product.

This is **Convenience Sampling** because participants are chosen simply because they are easy to reach.

Practice Exercises

Practice 1

A teacher writes the names of all 40 students on slips of paper, mixes them in a box, and randomly draws 8 names. What sampling technique is used?

Practice 2

A researcher wants to survey every 12th customer entering a supermarket after randomly choosing the first customer. What sampling technique is used?

Practice 3

A school has students from Grades 7, 8, 9, and 10. The researcher selects 10 students from each grade level. What sampling technique is used?

Practice 4

A city is divided into 15 barangays. The researcher randomly chooses 3 barangays and surveys every household in those barangays. What sampling technique is used?

Practice 5

A student surveys the first 20 classmates she meets during recess. What sampling technique is used?

Practice 6

Identify whether the following statement is about a **population** or a **sample**: “All Grade 7 students in the school.”

Practice 7

Identify whether the following statement is about a **population** or a **sample**: “Thirty Grade 7 students selected from the school.”

Practice 8

Which sampling technique gives every member of the population an equal chance of being selected?

Practice 9

Why is stratified sampling useful when the population consists of different groups?

Practice 10

Match each situation with the correct sampling technique.

label=a. Surveying every 8th visitor entering a museum.

lbbel=b. Drawing names from a box.

lcbel=c. Selecting all students from two randomly chosen sections.

ldbel=d. Surveying only nearby shoppers.

lebel=e. Choosing equal numbers of boys and girls.

Activities

Activity 1

A coach writes the names of all 30 players on cards, shuffles them, and randomly selects 6 players. What sampling technique is used?

Activity 2

A researcher surveys every 15th passenger entering a bus terminal after choosing a random starting passenger. What sampling technique is used?

Activity 3

A university has students from four colleges. The researcher selects 20 students from each college. What sampling technique is used?

Activity 4

A province is divided into several municipalities. The researcher randomly selects 4 municipalities and surveys all households in them. What sampling technique is used?

Activity 5

A vendor asks the first 25 customers who visit the store to answer a questionnaire. What sampling technique is used?

Activity 6

Identify whether the following statement is about a **population** or a **sample**: “All teachers in the district.”

Activity 7

Identify whether the following statement is about a **population** or a **sample**: “Forty teachers selected from the district.”

Activity 8

Which sampling technique involves selecting participants at regular intervals from a list?

Activity 9

Why is cluster sampling useful when the population is spread over a large geographic area?

Activity 10

Match each situation with the correct sampling technique.

label=a. Selecting every 5th patient arriving at a clinic.

lbbel=b. Randomly drawing employee ID numbers.

lcbel=c. Surveying all families in two selected villages.

ldbel=d. Asking only friends to answer a survey.

lebel=e. Selecting equal numbers of students from each grade level.

Thank you!